IVEY | Publishing

W43339

STARBUCKS DEEP BREW: AI-POWERED CUSTOMER EXPERIENCE[1]

Professor Derrick Neufeld wrote this case solely to provide material for class discussion. The author does not intend to illustrate either effective or ineffective handling of a managerial situation. The author may have disguised certain names and other identifying information to protect confidentiality.

This publication may not be transmitted, copied, digitized, used to train, input, or apply in a large language model or any other generative artificial intelligence tool, or otherwise reproduced in any form or by any means without the permission of the copyright holder. Reproduction of this material is not covered under authorization by any reproduction rights organization. To order copies or request permission to reproduce materials, contact Ivey Publishing, Ivey Business School, Western University, London, Ontario, Canada, N6G 0N1; (e) cases@ivey.ca; www.iveypublishing.ca. Please submit any errata to publishcases@ivey.ca.

Copyright © 2025, Ivey Business School Foundation *Version: 2025-06-26*

Starbucks Corporation (Starbucks) chief executive officer Kevin Johnson faced a pivotal decision: How could the company harness emerging technologies to enhance customer personalization and operational efficiency? Inspired by the McDonald's Corporation (McDonald's) acquisition of Dynamic Yield to bolster its artificial intelligence (AI) capabilities, Johnson recognized the need for Starbucks to innovate similarly. The result was the development of Deep Brew, an AI-driven platform designed to personalize customer experiences, optimize store operations, and manage inventory. The challenge was clear: Could Starbucks successfully integrate this advanced technology across its vast global network to meet evolving consumer expectations? With the company handling over 100 million transactions per week, leveraging AI to streamline operations and improve customer retention was not just an opportunity but a strategic imperative.[2]

The stakes in AI adoption were high. Across industries, AI-driven innovations transformed business operations. For example, Levi Strauss & Co. (Levi's) successfully used AI to analyze shopping trends and adjust its product offerings, leading to a 15 per cent increase in loose-fit jeans sales.[3] Similarly, luxury conglomerate LVMH Moët Hennessy Louis Vuitton SE (LVMH) established AI Factory to drive personalized experiences across its brands, proving that AI, when used effectively, could enhance customer engagement and operational efficiency.[4] But successful implementation was far from guaranteed. Forward Health, a healthcare start-up, attempted to revolutionize medical care with AI-powered CarePods, but the project collapsed due to technical failures and poor patient adoption.[5] Even in retail, AI was not foolproof—Shein's AI-driven fast fashion model led to accusations of environmental harm and unethical labour practices.[6]

Starbucks leadership understood that Deep Brew had the potential to become a transformative asset, but only if it was carefully implemented and vigilantly maintained.

INDUSTRY BACKGROUND

After experiencing steady growth for many years, the global coffee industry reached US$484 billion[7] in 2023 and was projected to expand at a compound annual growth rate (CAGR) of 5.2 per cent, to almost $700 billion, by 2030 (see Exhibit 1). Within this expanding market, specialty coffee grew even more rapidly, reaching a valuation of $101.6 billion in 2024, with expectations of a CAGR of 10.4 per cent from 2025 to 2030.[8] Consumers increasingly sought high-quality, ethically sourced craft coffee experiences, challenging the dominance of mass-market chains like Starbucks. This shift toward premiumization and sustainability forced industry leaders to innovate in sourcing, production, and customer engagement to meet evolving expectations.[9]

Do Not Copy or Post

At the same time, competition intensified, as third-wave coffee shops such as Blue Bottle Coffee Inc. (Blue Bottle), Stumptown Coffee Roasters (Stumptown), and Intelligentsia Coffee (Intelligentsia) focused on artisan roasting and direct-trade sourcing, appealing particularly to urban and millennial consumers.[10] Meanwhile, global competitors like Costa Coffee, Tim Hortons Inc., and Peet's Coffee aggressively expanded into new international markets, increasing pressure on Starbucks' dominance.[11] In addition, fast-food giants like McDonald's and Dunkin' Donuts LLC leveraged digital loyalty programs, AI-driven ordering, and competitive pricing to attract convenience-seeking customers, further reshaping the market landscape.[12]

Beyond competitive pressures, digital transformation reshaped the industry, with AI, Internet of Things (IoT), and cloud-based platforms playing an increasingly important role. Companies adopted AI-powered personalized promotions and recommendations, utilizing mobile data and purchase history to enhance customer experiences. Predictive analytics and demand forecasting helped optimize inventory management and reduce waste, while automated coffee machines and robotic baristas, such as those in Cafe X, streamlined operations and improved efficiency.[13]

Sustainability and ethical sourcing also emerged as critical industry drivers, with consumers demanding greater transparency, carbon-neutral initiatives, and fair trade certification. Major coffee retailers responded by integrating traceability technologies and reinforcing sustainable supply chain practices to align with shifting consumer values.[14]

The coffee industry reached an inflection point—while growth opportunities abounded, evolving consumer preferences, technological disruptions, and increased competition challenged traditional business models. The Starbucks Deep Brew initiative emerged as a strategic response to these industry dynamics, aiming to leverage AI and data analytics to maintain a competitive advantage and redefine customer engagement in a rapidly shifting market.[15]

COMPANY BACKGROUND

Founded in 1971 in Seattle, Starbucks grew into the largest coffee chain in the world, operating more than 38,000 locations across over 80 markets[16] with revenue in excess of $36 billion in 2024 (see Exhibit 2). The company established itself in a unique position between premium coffee shops and fast-food coffee providers, offering high-quality beverages alongside digital convenience. Unlike independent specialty coffee shops, which emphasized artisanal quality, or fast-food competitors, which focused on speed and price, Starbucks balanced quality, customer experience, and technological innovation to sustain its market dominance.[17]

A key driver of Starbucks' success was its strong brand loyalty. The Starbucks Rewards program exceeded 31 million active members, contributing nearly 57 per cent of US revenue, making it one of the most effective loyalty programs in the food and beverage industry.[18] Starbucks also gained an early advantage in digital transformation, pioneering mobile payments and ordering well before its competitors. By 2019, nearly 30 per cent of total sales comprised digital transactions, reinforcing its position as a tech-savvy retailer.[19] Additionally, the company exercised direct control over its supply chain, sourcing coffee beans through its C.A.F.E. (Coffee and Farmer Equity) Practices. This approach ensured ethical sourcing, sustainable production, and strong relationships with farmers, giving Starbucks a competitive edge in an eco-conscious market.[20]

Despite its strengths, Starbucks faced significant competitive challenges. The rise of specialty coffee brands such as Blue Bottle, Stumptown, and Intelligentsia introduced a more curated, high-end experience, appealing to consumers willing to pay a premium for quality and sustainability. Additionally, labour costs and operational efficiency pressures forced Starbucks to consider greater automation and AI-driven

Do Not Copy or Post

scheduling to manage rising expenses. Expanding globally also presented challenges, as Starbucks had to adapt to local preferences and economic fluctuations, making it difficult to standardize its business model while maintaining the authenticity of regional offerings.

DIGITAL EVOLUTION

Starbucks consistently positioned itself as a technology-driven retailer, making significant investments in AI, data analytics, and cloud computing. Johnson, who led the company's digital transformation before stepping down in 2022, famously viewed Starbucks as a tech company as much as a coffee brand. Under his leadership, the company built AI-powered analytics to personalize customer interactions, optimize store operations, and enhance supply chain efficiencies. His successor, Laxman Narasimhan, continued this focus, emphasizing automation and AI integration as key drivers of Starbucks' future growth.[21]

Starbucks AI investments were aligned with three core strategic priorities. First, the company leveraged AI-powered insights to enhance personalization at scale, tailoring customer experiences both digitally and in-store. Second, Starbucks automated scheduling, inventory forecasting, and predictive maintenance through machine learning models, significantly improving operational efficiency. Finally, AI played a pivotal role in customer engagement, allowing Starbucks to boost loyalty, drive repeat business, and increase digital ordering adoption through AI-driven recommendations and gamified rewards.[22]

With this technological foundation in place, the introduction of Deep Brew was a natural next step in Starbucks' digital transformation journey. By combining AI-powered decision-making with customer data and predictive analytics, Deep Brew promised to enhance operational efficiency, improve personalization, and create new opportunities for market growth. However, its success depended on seamless integration, customer adoption, and the company's ability to navigate potential ethical and operational challenges.[23]

DEEP BREW TECHNOLOGY AND DATA INFRASTRUCTURE

Starbucks' data collection strategy was multifaceted, leveraging various touchpoints to gather comprehensive customer insights. The Starbucks mobile app, launched in 2011, served as a primary data source, enabling customers to place orders, make payments, and participate in the loyalty program. This platform captured detailed information on purchase history, preferences, and visit frequency. With over 31 million active members as of 2023, Starbucks Rewards became a powerful driver of sales, contributing nearly 57 per cent of the company's revenue in the US.[24]

However, Starbucks' data collection extended far beyond digital interactions. In-store IoT-enabled equipment, such as the Mastrena espresso machines, collected data on beverage preparation and machine performance. These high-tech espresso machines were equipped with sensors that tracked metrics such as temperature, pressure, and maintenance needs, allowing the company to maintain consistent beverage quality while predicting and preventing breakdowns.[25]

Deep Brew AI infrastructure relied on the Microsoft Azure cloud-based computing platform, allowing for seamless integration and scaling of machine learning models.[26] Azure provided the infrastructure to handle billions of data points across Starbucks' digital and physical touchpoints, processing real-time insights into customer behaviour, sales trends, and store operations. For example, external data sources, such as weather conditions and local events, provided contextually relevant customer experiences. If a sudden heat wave was forecasted, for example, Deep Brew might automatically adjust store inventory to ensure an ample supply of iced beverages, reducing waste and maximizing sales. The AI models housed within Azure were continuously updated to refine recommendations and improve forecasting accuracy.[27]

Do Not Copy or Post

Starbucks' enterprise data analytics platform (EDAP) functioned as a centralized data repository, integrating information from multiple sources, including mobile orders, in-store transactions, Wi-Fi analytics, loyalty program interactions, social media monitoring, foot traffic patterns, dwell times, popular areas within stores, store layout optimization, product placement, and more.[28] These data streams fed into machine learning models that powered dynamic menu boards, automated promotions, and real-time inventory adjustments. For example, digital drive-through menus used AI to customize recommendations based on weather, time of day, and customer purchase history. If a customer frequently ordered iced coffee in the afternoon, Deep Brew might suggest a cold brew upon their arrival, with an additional discount if the AI predicted lower-than-expected sales for that item. Similarly, Starbucks could tailor location-based promotions, offering discounts on breakfast sandwiches to customers within a store's Wi-Fi radius who had a history of purchasing food items. The EDAP allowed Starbucks to break down data silos so that different departments—marketing, operations, and supply chain—could leverage the same real-time insights for decision-making. Real-time data analysis allowed Starbucks to adjust promotional campaigns dynamically, targeting specific customer segments based on location, time of day, and seasonal preferences.[29]

Social media and market research data were also instrumental. Actively monitoring social media platforms allowed Starbucks to gauge customer sentiment, identify emerging trends, and respond to feedback in real time. By tracking mentions, hashtags, and reviews, Starbucks could detect shifts in customer preferences and tailor its marketing campaigns accordingly. Social media data could also be used to personalize digital offers. For instance, customers who frequently posted about trying seasonal beverages might receive early access to limited-time menu items.[30] Market survey data likewise provided direct insights into consumer preferences, product feedback, and brand perceptions. This qualitative data complemented existing quantitative data, offering a holistic view of customer needs and areas for improvement. Starbucks could use survey data to test new product concepts or measure customer satisfaction with recent changes to the in-store experience.[31]

Automated staffing, inventory management, and related internal operations were further optimized by Deep Brew. Starbucks used historical transaction data and AI-driven forecasting models to determine staffing needs and help store managers schedule shifts more effectively. By predicting peak hours, the system prevented overstaffing during slow periods and ensured adequate coverage during rushes, improving labour efficiency while reducing costs.[32] In inventory management, AI algorithms tracked product consumption rates and automatically adjusted supply chain orders. If a specific store was experiencing a surge in matcha latte orders due to a viral social media trend, Deep Brew could adjust replenishment schedules to ensure that inventory remained balanced. This automated approach reduced waste, prevented stockouts, and increased operational efficiency and profitability.[33]

To ensure the protection of customer data, Starbucks implemented a comprehensive data governance framework that adhered to global privacy regulations, including the General Data Protection Regulation (GDPR) and the California Consumer Privacy Act (CCPA). This framework encompassed measures such as encrypting personal data and anonymizing sensitive information to safeguard customer privacy. Additionally, Starbucks provided customers with control over their privacy settings through the Starbucks mobile app, allowing them to manage their personal information and communication preferences. These initiatives underscored Starbucks' commitment to data transparency and security, fostering trust between the company and its customers.[34]

"Deep Brew will increasingly power our personalization engine, optimize store labour allocations, and drive inventory management in our stores," Johnson stated, highlighting the strategic importance of data-driven decision-making in the company's future.[35] By ensuring transparency and security in its data management processes, Starbucks aimed to balance personalization with privacy, maintaining consumer trust while delivering customized experiences.

Do Not Copy or Post

FUTURE CHALLENGES

Companies like McDonald's, Levi's, and LVMH showcased the transformative potential of AI when implemented effectively, but some other firms encountered extremely serious challenges. Forward Health's AI-driven CarePods collapsed due to operational inefficiencies, technical failures, and low customer adoption. Shein's AI-powered fast fashion model, despite its operational success, sparked widespread concerns over environmental harm and unethical labour practices. As Starbucks scaled Deep Brew, it could not afford complacency. Ensuring responsible AI deployment and proactively addressing potential pitfalls would be essential for long-term success.

Deep Brew had firmly established Starbucks as a leader in AI-driven customer engagement, but its continued operation presented ongoing challenges. One persistent issue was maintaining seamless technological integration across thousands of stores worldwide. Unlike digital-first companies, Starbucks operated in a complex physical retail environment where AI systems needed to work in tandem with human employees. Ensuring that store managers and baristas could effectively interpret and act on AI-generated insights remained critical to maximizing Deep Brew's potential. Rather than replacing human expertise, automation needed to enhance decision-making and operational efficiency. Additionally, sustained adoption depended on overcoming resistance to change, requiring Starbucks to continually refine its training programs and change management strategies to ensure long-term alignment between AI-driven processes and frontline execution.

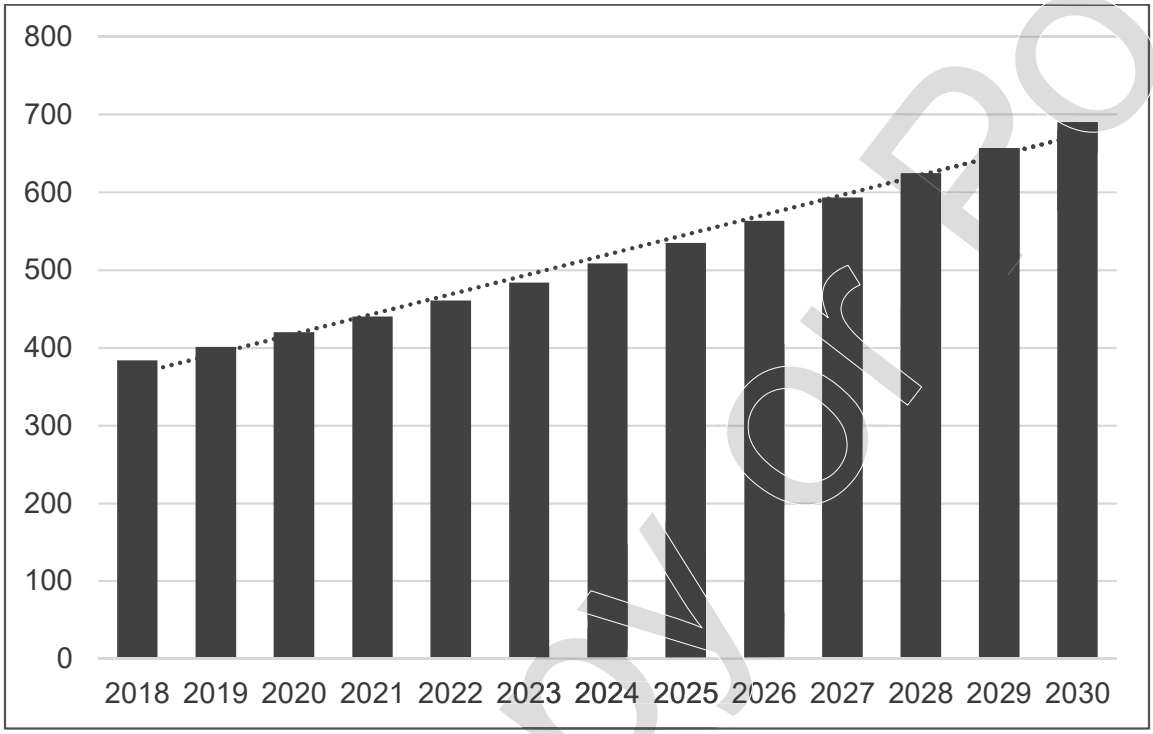
Another key concern was data privacy. As AI-driven personalization advanced, customers grew wary of how their data was collected and used. Any perceived overreach—such as excessive tracking or unclear consent—could trigger backlash, harming Starbucks' reputation and consumer loyalty. With evolving regulations like GDPR and CCPA, compliance required proactive policy updates, stronger encryption, and greater transparency. Balancing personalization with privacy remained essential to maintaining trust and avoiding regulatory risks.

AI-driven decision-making also raised ethical concerns about algorithmic bias. If Deep Brew primarily promoted high-margin products, it could disadvantage certain customer segments or limit product diversity by reinforcing existing purchasing patterns. Overreliance on AI recommendations might also reduce exposure to new or seasonal offerings. To ensure fairness and inclusivity, Starbucks needed to refine its algorithms continuously, balancing profitability with customer choice while maintaining a diverse and engaging product experience.

Johnson's decision to invest in AI reflected a broader shift in retail, where data-driven personalization had become a competitive necessity. Companies that seamlessly integrated AI into their operations stood to gain a significant edge, while those that lagged risked obsolescence. As Starbucks refined its AI strategy, its successes and challenges would offer valuable lessons—as both a blueprint for innovation and a cautionary tale for businesses navigating the complexities of AI-powered retail.

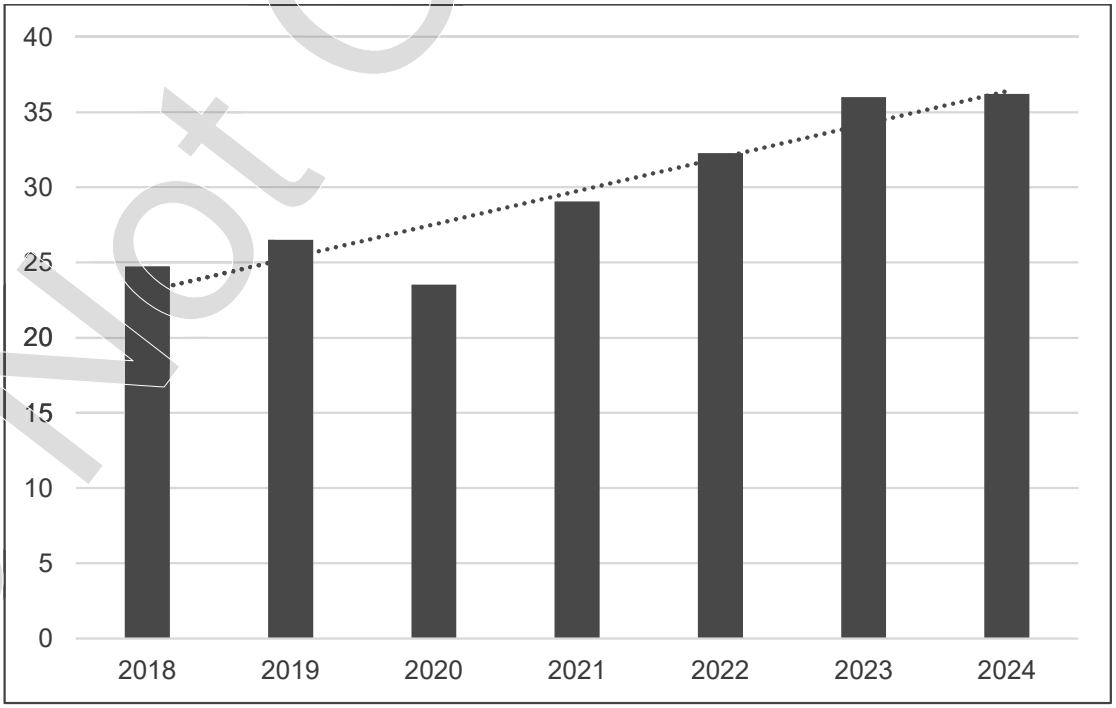
Do Not Copy or Post

EXHIBIT 1: GLOBAL COFFEE MARKET (US$ BILLION)



Source: Created by the case author based on "Global Coffee Market Size & Outlook, 2023–2030," Grand View Research, accessed February 6, 2025, https://www.grandviewresearch.com/horizon/outlook/coffee-market-size/global.

EXHIBIT 2: STARBUCKS REVENUE AND GROWTH (US$ BILLION)



Source: Created by the case author based on "Revenue of Starbucks from 2003 to 2024," Statista, accessed February 6, 2025, https://www.statista.com/statistics/266466/net-revenue-of-the-starbucks-corporation-worldwide.

Do Not Copy or Post

ENDNOTES

[1] This case has been written on the basis of published sources only. Consequently, the interpretation and perspectives presented in this case are not necessarily those of Starbucks or any of its employees.
[2] "Deep Brew: Transforming Starbucks into AI & Data-Driven Company," Hyperight, June 30, 2021, https://hyperight.com/deep-brew-transforming-starbucks-into-a-data-driven-company.
[3] Isabelle Bousquette, "How Tech Helped Levi's Ride the 'Baggy Jeans' Trend," *Wall Street Journal*, January 27, 2025, https://www.wsj.com/articles/how-tech-helped-levis-ride-the-baggy-jeans-trend-f290721d.
[4] Maghan McDowell, "Inside LVMH's AI Factory," Vogue Business, July 9, 2024, https://www.voguebusiness.com/story/technology/inside-lvmhs-ai-factory.
[5] "Inside Forward's Failed Attempt to Revolutionize the Doctor's Office," Business Insider, November 13, 2024, https://www.businessinsider.com/healthcare-startup-forward-shutdown-carepod-adrian-aoun-2024-11.
[6] Sachi Mulkey, "AI Has Helped Shein Become Fast Fashion's Biggest Polluter," Wired, September 14, 2024, https://www.wired.com/story/shein-is-officially-the-biggest-polluter-in-fast-fashion-ai-is-making-things-worse.
[7] All dollar amounts are in US dollars.
[8] Research and Markets, "Specialty Coffee Market Size, Share & Trends Analysis Report and Segment Forecasts, 2025–2030," GlobeNewswire, November 19, 2024, https://www.globenewswire.com/news-release/2024/11/19/2983848/28124/en/Specialty-Coffee-Market-Size-Share-Trends-Analysis-Report-and-Segment-Forecasts-2025-2030.html.
[9] "Global Coffee Market Size & Outlook, 2023–2030," Grand View Research, accessed February 6, 2025, https://www.grandviewresearch.com/horizon/outlook/coffee-market-size/global.
[10] Third Wave Coffee, "How Third Wave Coffee Roasters Are Redefining Coffee Culture," Lux Cafe Club, May 8, 2024, https://www.luxcafeclub.com/blogs/news/how-third-wave-coffee-roasters-are-redefining-coffee-culture.
[11] "Starbucks' Top Competitors: Identifying Key Challengers," Subscribed.fyi, April 23, 2024, https://subscribed.fyi/blog/starbucks-top-competitors-identifying-key-challengers.
[12] "Global Cafe And Bars Market Overview," Market Research Future, accessed February 6, 2025, https://www.marketresearchfuture.com/reports/cafe-bars-market-26218.
[13] "Commercial Products," Cafe X, accessed February 6, 2025, https://www.cafexapp.com/commercial.
[14] "Coffee," Starbucks, accessed February 6, 2025, https://www.starbucks.com/responsibility/sourcing/coffee; "Recognition of Fairtrade More Than Doubles in Four Years, per 2023 Consumer Market Research," Fairtrade America, September 7, 2023, https://fairtrade.net/us-en.html.
[15] "Deep Brew."
[16] "Number of Starbucks Stores Worldwide from 2003 to 2024," Statista, December 10, 2024, https://www.statista.com/statistics/266465/number-of-starbucks-stores-worldwide.
[17] "About Us," Starbucks, accessed February 6, 2025, https://www.starbucks.com/about-us.
[18] Bryan Wassel, "Starbucks to Capitalize on Record Loyalty Membership with App Improvements," CX Dive, February 1, 2024.
[19] Taylor Soper, "Starbucks Mobile Orders Surpass 30% of Total Transactions at U.S. Stores," GeekWire, January 31, 2024, https://www.geekwire.com/2024/starbucks-mobile-orders-surpass-30-of-total-transactions-at-u-s-stores-for-the-first-time.
[20] "Starbucks Reports Q4 and Full Year Fiscal 2019 Results," Starbucks, October 30, 2019, https://investor.starbucks.com/news/financial-releases/news-details/2019/Starbucks-Reports-Q4-and-Full-Year-Fiscal-2019-Results/default.aspx.
[21] "Starbucks Announces Triple Shot Reinvention Strategy with Multiple Paths for Long-Term Growth," Starbucks, November 2, 2023, https://about.starbucks.com/press/2023/starbucks-announces-triple-shot-reinvention-strategy-with-multiple-paths-for-long-term-growth.
[22] "Case Study: Starbucks Revolutionizes the Coffee Experience with AI," AI Expert Network, February 7, 2024, https://aiexpert.network/case-study-starbucks-revolutionizes-the-coffee-experience-with-ai.
[23] "Deep Brew."
[24] "Starbucks Reports Q4 and Full Year Fiscal 2023 Results," Starbucks, November 2, 2023, https://investor.starbucks.com/news/financial-releases/news-details/2023/Starbucks-Reports-Q4-and-Full-Year-Fiscal-2023-Results/default.aspx.
[25] James Bourne, "Starbucks Flips the Equation with Deep Brew – By Using AI For a More Humanised Customer Experience," Marketing Tech, November 4, 2019, https://www.marketingtechnews.net/news/deep-brew-starbucks-aims-use-ai-more-humanised-customer-experience.
[26] Samantha Schwartz, "Starbucks' AI Project Is Turning Past Investments into Data Insights," CIO Dive, November 1, 2019, https://www.ciodive.com/news/starbucks-ai-project-is-turning-past-investments-into-data-insights/566362.
[27] "It's Coffee Time: Find Out How Starbucks Turns to Technology to Brew Up a More Personal Connection with Its Customers," Microsoft, June 14, 2019, https://blogs.microsoft.com/conexiones/2019/06/14/its-coffee-time-find-out-how-starbucks-turns-to-technology-to-brew-up-a-more-personal-connection-with-its-customers.
[28] "Deep Brew."
[29] "How Starbucks Is Using AI and More to Better Serve Its Customers," Future Stores, accessed February 6, 2025, https://futurestores.wbresearch.com/blog/starbucks-ai-serve-customers-strategy.
[30] "Unveiling Starbucks Social Media Marketing Strategy and Key Takeaways," Radarr, accessed February 6, 2025, https://www.radarr.com/blog/unveiling-starbucks-social-media-marketing-strategy.
[31] "Starbucks Market Research Strategy: What It Is & Why It Works," Drive Research, February 10, 2023, https://www.driveresearch.com/market-research-company-blog/starbucks-market-research-strategy.
[32] "Deep Brew."
[33] "Big Data: The Secret to Starbucks' Supply Chain Success," Sisense, March 21, 2024, https://www.sisense.com/blog/big-data-the-secret-to-starbucks-supply-chain-success-2.
[34] "Starbucks Privacy Notice," Starbucks, January 17, 2025, https://www.starbucks.com/terms/privacy-notice.
[35] "Big Data."

Do not copy or Post